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# UNIFAC AND RELATED GROUP-CONTRIBUTION MODELS FOR PHASE EQUILIBRIA

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## ABSTRACT

The advent of the ASOG and UNIFAC group-contribution methods for the prediction of activity coefficients approximately 15 years ago was a significant boost to the ability of chemical engineers to model chemical processes. This paper reviews the status and recent progress in the UNIFAC group-contribution model for predicting phase equilibria.

Several different types of liquid mixtures are considered: Non-electrolyte mixtures with normal boiling components, mixtures with polymers, and mixtures including strong electrolytes. It is concluded that in recent years much progress has been made in the development of group-contribution models for the prediction of activity coefficients. It is in addition shown that combining activity coefficient models with equations of state renders the group-contribution approach applicable also to the high pressure region, including gas solubility.

## THE GAMMA-PHI AND THE EOS APPROACH

The condition of equilibrium in a closed system with $N$ components at temperature $T$ and pressure $P$ is that the Gibbs function is minimum. Classical thermodynamics shows that for vapor($V$) - liquid ($L$) equilibrium this condition can be rewritten as follows:

$$f_i^V = f_i^L \qquad i = 1,2 \ldots N \tag{1}$$

where $f_i$ is the fugacity of component $i$ in the mixture.

This equation may be rewritten in terms of the vapor-phase fugacity coefficient $\Phi_i^V$ and mole fraction $y_i$ and the liquid-phase activity coefficient $\gamma_i$, mole fraction $x_i$, and standard state fugacity $f_i^o$.

$$\Phi_i^V y_i P = x_i \gamma_i f_i^o \tag{2}$$

The standard state fugacity $f_i^o$ is the fugacity of pure component $i$ at the temperature and pressure of the system. Hence $\gamma_i$ is unity for a pure component $i$.

At low to moderate pressures, $f_i^o$ is often equal to the pure component vapor pressure at the system temperature, $P_i^{sat}$. In addition $\Phi_i^V$ is under these conditions often equal to unity. Hence equation (2) becomes:

$$y_i P = x_i \gamma_i P_i^{sat} \tag{3}$$

In this case the calculation of equilibrium ratios for VLE is dependent on the availability of a reliable method for the estimation of $\gamma_i$. Due to the assumptions involved, this approach is only applicable to mixtures at relatively low pressures.

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At high pressures the equation of state (EOS) approach is used for the computation of VLE. Equation (1) is in this case rewritten as follows:

$$\Phi_i^V y_i = \Phi_i^L x_i \tag{4}$$

The fugacity coefficients $\Phi_i^L$ and $\Phi_i^V$ are related to temperature, pressure and composition using established thermodynamic relationships.

Traditionally, equation (2) is used for VLE computations for low pressure systems and mixtures with both polar and non-polar components. The advantage is simplicity and coverage of a large variety of mixtures of interest in chemical technology. A major disadvantage is the pressure limitation.

On the other hand, equation (4) is traditionally used only for mixtures with non-polar components (mainly hydrocarbons). The advantage of this approach is that it applies to both low and high pressures. The disadvantage is the relatively limited range of components for which the approach applies.

Today, the two approaches are being merged into one single model, thus taking advantage of the positive features of both approaches. The only disadvantage is that of additional computer time requirements.

## THE UNIFAC ACTIVITY COEFFICIENT MODEL

Gmehling et al. (1988), and Heidemann and Fredenslund (1987) have recently reviewed group-contribution models for the prediction of activity coefficients. It will not be attempted here to repeat these reviews, but rather to give status of the UNIFAC model and its modifications.

The last decade has seen a nearly explosive growth in the application of the UNIFAC group-contribution model (Fredenslund et al., 1977, Tiegs et al.,1987). The applications are in areas such as design of chemical separation processes (see e.g. Torres-Marchal and Cantalino, 1986), in flash point calculations (Wu et al., 1988 and Visser, 1988), in estimation of environmental exposures (see e.g. Arbucle, 1986), and many others. The reason for this growth is the reliability and availability of a larger number of UNIFAC group-interaction parameters.

In the UNIFAC model the activity coefficient is given in terms of a combinatorial contribution taking into account entropy effects arising from differences in molecular size and shape, and a residual contribution taking into account energetic interactions between the functional groups in the mixture. This is shown schematically in Figure 1:

$$G^E = H^E - TS^E$$

$$\ln \gamma_i = \ln \gamma_i^R + \ln \gamma_i^C \quad ; \quad \ln \gamma_i^C = f(x_i, q_i, r_i)$$

$$q_i = \sum \upsilon_k^{(i)} Q_k \quad \textit{relative van der Waals area}$$

$$r_i = \sum \upsilon_k^{(i)} R_k \quad \textit{relative van der Waals volume}$$

$R_k$ *and* $Q_k$ *are given by Bondi* (1968)

$$\ln \gamma_i^R = \sum \upsilon_k^{(i)} \left[ \Gamma_k - \Gamma_k^{(i)} \right] \quad ; \quad \Gamma_k = f(x_i, q_i, a_{mn})$$

**Figure 1.** The UNIFAC model.

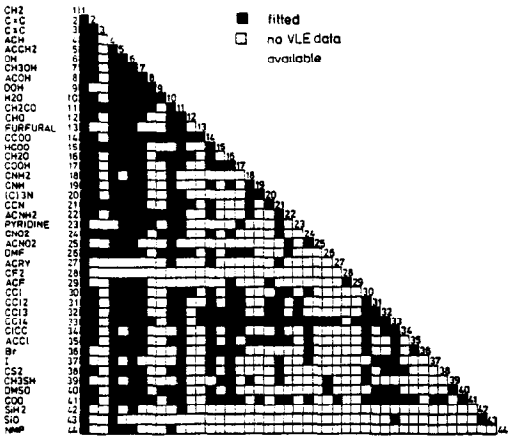



**Figure 2.** Present status of the UNIFAC parameter matrix.

In Figure 1, $\Gamma_k$ and $\Gamma_k^{(i)}$ are the group activity coefficients, and $a_{mn}$ the group-interaction parameters. Figure 2 shows the groups and the availability of group-interaction parameters for UNIFAC predictions of vapor-liquid equilibria (UNIFAC-VLE).

Limitations of UNIFAC

In spite of its many successful applications the UNIFAC model does have its limitations. Very briefly, they are the following:

1. Due to the solution-of-groups assumption, mixtures containing isomers cannot be treated using UNIFAC.
2. The $\gamma$-approach (equation (2)) limits UNIFAC to applications below 10-15 atm. in pressure.
3. The temperature range is limited to temperatures approximately between 275 and 425 K.
4. Non-condensable gases are not included.
5. Electrolytes (such as NaCl) are not included.
6. Polymers are not included.
7. It has been found that UNIFAC with parameters based on VLE data (UNIFAC-VLE) does not yield quantitative prediction of LLE. This has led Magnussen et al. (1981) to publish a UNIFAC-LLE parameter table especially aimed at predicting LLE at 25° C. The UNIFAC-LLE model is identical to UNIFAC-VLE, but the parameters differ. This indicates a fundamental deficiency of the solution-of-groups approach - the activity coefficients do not "know" what they are being used for, and hence in principle it should be possible to use the same model with the same parameters for all applications.

It is questionable if anything can be done about limitations no. 1 and 7. However, efforts to overcome limitations no. 2-6 will be described below.

Modifications of UNIFAC

In the original UNIFAC model the group-interaction parameters are assumed to be independent of temperature. For this reason the original UNIFAC model does not yield quantitative predictions of the excess enthalpy, $H^E$.

Independently, modifications of the UNIFAC model have been developed (Gmehling and Weidlich, 1986, and Larsen et al., 1987), encompassing modification of both the combinatorial and residual parts of the model. The modifications of the combinatorial part will be described in the section below, and the modification of the residual part consists of making the group-interaction parameter, $a_{mn}$, temperature dependent. This enables quantitative simultaneous correlation of VLE and $H^E$ data, which again assures good temperature dependence of the activity coefficients in most cases up to 550-600 K.

The above-mentioned two modified versions of the UNIFAC model are slightly different in detail. The two different versions will be merged into one version.

For the modified UNIFAC model of Gmehling and Weidlich (1986), Gmehling et al. (1988) report the following results:

| | Number of data sets | Deviation | | |
|---|---|---|---|---|
| | | | Original UNIFAC | Modified UNIFAC |
| Binary systems | | | | |
| VLE | 7690 | $\lvert\Delta y\rvert$ | 0.010 | 0.008 |
| $H^E$ | 12360 | $\lvert\Delta H^E/H^E_{max}\rvert$ | 42% | 13% |
| Ternary systems | | | | |
| VLE | 3457 | $\lvert\Delta y\rvert$ | 0.015 | 0.013 |
| $H^E$ | 1478 | $\lvert\Delta H^E/H^E_{max}\rvert$ | 51% | 11% |

**Table 1.** Deviation between experimental and predicted vapor-phase compositions and excess enthalpies using the original and the modified UNIFAC models.

Similar improvements are reported by Larsen et al. (1987). The conclusion is that a modified UNIFAC does not offer much improvement over original UNIFAC for VLE predictions. However, due to the temperature dependence of the interaction parameters, the prediction of $H^E$ - and hence the temperature dependence of the activity coefficients - is much improved. Thus large temperature extrapolations are safer with the modified UNIFAC than with the original UNIFAC model.

The Combinatorial Term

In the group-contribution models used today for VLE calculation, the energy interaction parameters between all alkane groups ($CH_3$, $CH_2$, CH, and C) are assumed equal to zero. From a pragmatic point of view, then the role of the combinatorial term is to yield as good predictions as possible of activity coefficients in mixtures containing only alkanes. Some combinatorial expressions are:

$$ASOG:\ \ln \gamma_i^C = 1 - \Phi_i/x_i + \ln \Phi_i/x_i \qquad (Flory - Huggins) \tag{5}$$

$$UNIFAC:\ \ln \gamma_i^C = 1 - \Phi_i/x_i + \ln \Phi_i/x_i - zq_i(1 - \Phi_i/\theta_i + \ln\ \Phi_i/\theta_i) \tag{6}$$

$$(Stavermann - Guggenheim)$$

$$\text{Mod. UNIFAC (Dortmund)}: \ln \gamma_i^C = 1 - \Phi_i'/x_i + \ln \Phi_i'/x_i - zq_i(1 - \Phi_i'/\theta_i + \ln \Phi_i'/\theta_i) \quad (7)$$

$$\Phi_i' = r_i^{3/4}/\sum r_j^{3/4} x_j$$

$$\text{Mod. UNIFAC (Lyngby)}: \ln \gamma_i^C = 1 - \Phi_i''/x_i + \ln \Phi_i''/x_i \quad (8)$$

$$\Phi'' = r_i^{2/3}/\sum r_j^{2/3} x_j$$

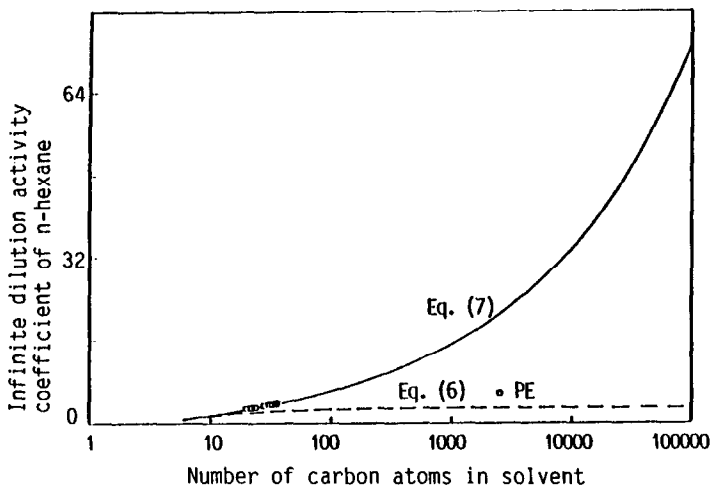



**Figure 3.** Activity coefficient of hexane infinitely diluted in n-alkanes as a function of the number of carbon atoms in the solvent (the activity coefficient is on weight basis, PE is polyethylene).

Figure 3 shows predictions of the activity coefficient of hexane infinitely diluted in other n-alkanes. The carbon number of the solvent alkane is given on the abscissa. It may be seen that the predicted infinite dilution activity coefficients using equation (6) are too small, whereas the empirical modifications given by equations (7) (and (8)) give the expected improvements for mixtures with low molar mass components. Interesting enough, it may also bee seen that equation (8) does not extrapolate well to mixtures with polyethylene, and hence this combinatorial should not be used for mixtures with polymers.

As shown in Figure 1, the UNIFAC activity coefficient has an entropic and an enthalpic ("residual") contribution. Elbro et al. (1988) have derived a new expression for the entropic activity coefficient, $\gamma^{ent}$, from the generalized van der Waals partition function. The expression is quite similar to equation (5) but includes both a combinatorial and a free volume contribution:

$$\ln \gamma_i^{ent} = 1 - \Phi_i^{fv}/x_i + \ln \Phi_i^{fv}/x_i \quad (9)$$

where

$$\Phi_i^{fv} = x_i(v_i - v_i^*)/\sum x_j/(v_j - v_j^*) \quad (10)$$

and $v^*$ is the molar hardcore volume calculated from the van der Waals volumes as in UNIFAC and $v$ is the molar volume. The total free volume, $v_f$, is here assumed to be what Bondi labelled the empty volume, i.e. $v_f = n\,(v\text{-}v^*)$.

The free volume effect may be quantified by looking at the difference between $ln\ \gamma_i^{ent}$ given by equation (9) and $ln\ \gamma_i^{C}$ given by the Flory-Huggins equation. The result is:

$$ln\,\gamma_i^{ent} - ln\,\gamma_i^{C} = ln\,\frac{(FV\%)_i}{(FV\%)_{mix}} + \frac{v_i}{v_{mix}}\left[1 - \frac{(FV\%)_i}{(FV\%)_{mix}}\right] \qquad (11)$$

where $(FV\,\%)_i$, $(FV\%)_{mix}$ denote free volume % for a pure component and a mixture respectively, and are given by

$$(FV\%)_i = \left(\frac{v_i - v_i^*}{v_i}\right)100 \qquad (FV\%)_{mix}\left(\frac{v_{mix} - v_{mix}^*}{v_{mix}}\right)100 \qquad (12,13)$$

From equation (11) it can be seen that the free volume contribution to the activity coefficient can only be ignored if all components in a mixture have equal free volume percentages. Free volume effects are normally only taken into account for mixtures with polymers.

In Figure 4, however, the free volume percent, $(FV\,\%)_i$, for approximately 20 polymers and 40 common solvents at $25^0$ C is plotted against the specific volume. The free volume percent of solvents ranges between 40 and 50 excluding water which has a free volume percent of 36. Polymers, on the other hand, generally have a free volume percent between 30 and 40. This demonstrates on a quantitative basis that polymers and solvents have very different free volume percentages.

But Figure 4 also shows that even for mixtures of low-molecular-weight solvent systems, free volume differences can reach the same magnitude as in polymer solutions. Consequently, if a free volume contribution is necessary in order to describe phase equilibria for polymer solutions, it might also advantageously be included in models for solvent mixtures. The UNIFAC activity coefficient model has been derived under the assumption that the free volume percent is constant, independent of component and composition. Figure 4 shows that this assumption is incorrect; hence some free volume effects have been incorporated in the fitted group-interaction parameters.

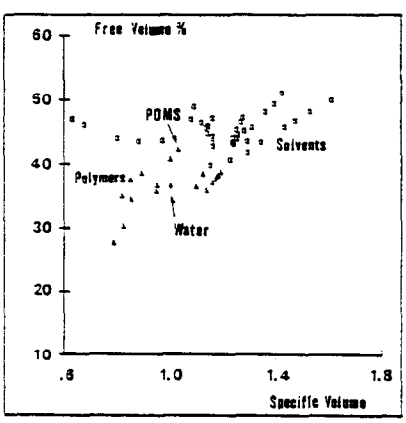



Figure 4. Calculated free volume % for amorphous polymers (Δ)and solvents (□) (including water) at $25^0$ C

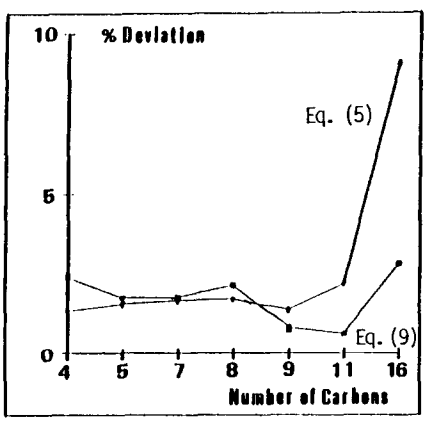



Figure 5. Mean percent deviations between calculated and experimental vapor pressures for hydrocarbon mixtures. n-hexane + n-$C_nH_{2n+2}$.

In Figure 5 the deviations between calculated and experimental vapor pressures are shown for a series of hydrocarbon mixtures. The activities have been calculated by the two forms of the Flory-Huggins combinatorial, equation (5), and by the new entropic activity coefficient, equation (9). For mixtures where the components are similar in size and free volume differences, the two equations result in nearly the same deviations between the calculated and experimental pressures. But as the size and free volume differences increase there is a marked improvement in using equation (9).

This indicates that it might be advantageous to replace the combinatorials of the UNIFAC and of the modified UNIFAC with equation (9). This is presently being investigated.

The use of the entropic activity coefficient expression in connection with polymer solutions will be discussed in a subsequent section.

UNIFAC and Surface Tensions

Under certain assumptions the surface tension of a liquid mixture, $\sigma$, is related to the surface tension of pure component i, $\sigma_i$, as follows:

$$\sigma = \sigma_i + (RT/A_i) \ln (x_{i,s} \gamma_{i,s} / x_{i,b} \gamma_{i,b}) \qquad i = 1,2...N \tag{14}$$

$$(\sum x_{i,s} = 1)$$

$A_i$ is the molar surface area of pure component $i$, $x_{i,s}$ and $x_{i,b}$ are the mole fractions in the surface and bulk liquid phases, and $\gamma_{i,s}$ and $\gamma_{i,b}$ are the corresponding activity coefficients.

Suarez et al. (1988) used the modified UNIFAC (Larsen et al., 1987), to predict $\gamma_{i,s}$ and $\gamma_{i,b}$. Pure component values for $\sigma_i$ and $A_i$ are determined from selected experimental surface tension data for binary mixtures. They obtained excellent agreement between experimental and calculated surface tensions.

An example of the results of Suarez et al. (1988) is given in Figure 6.

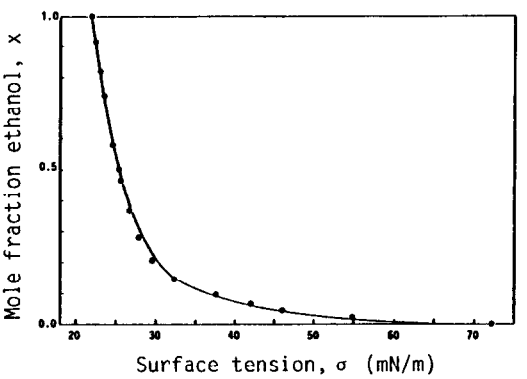



**Figure 6.** Experimental and calculated surface tensions of ethanol/water.

Wu (1986) has shown how an UNIFAC type approach may be used to predict the viscosities of liquid mixtures.

Infinite Dilution Activity Coefficients, $\gamma_i^\infty$

Bastas et al. (1988) indicate that for 8301 $\gamma_i^\infty$ data points ( ! ), the mean deviation between experimental values and those calculated from the original UNIFAC-VLE model is 28%. When they refit the UNIFAC parameters to $\gamma_i^\infty$ data only, they obtain a mean deviation of 20%, which is a surprisingly marginal improvement.

Gmehling et al. (1988), on the other hand, find that for a limited range of systems and with their modified UNIFAC model, significant improvement in the

prediction of $\gamma_i^\infty$ may be obtained. Similar results are found with the MOSCED model (Thomas et al., 1984).

It is in the opinion of the author that careful reconsiderations of the combinatorial (or entropic) contribution, coupled with care in the parameter estimation procedure, would result in a UNIFAC-VLE model with significantly improved predictions of $\gamma_i^\infty$ without loss of accuracy in the more concentrated region.

## UNIFAC AT HIGH PRESSURES: THE UNIWAALS EQUATION OF STATE

The van der Waals equation of state is defined by the following expression for the residual Helmholtz energy, $A^R$:

$$\frac{A^R}{RT} = -ln(1-\frac{b}{v}) - \frac{a(T)}{RTv} \tag{15}$$

The pressure and the excess Gibbs function resulting from this model are:

$$P = \frac{RT}{v-b} - \frac{a}{v^2} \tag{16}$$

and

$$\frac{G^E}{RT}|_{EOS} = -ln(1-\frac{b}{v}) - \frac{a}{RTv} + \sum_i x_i ln(1-\frac{b_i}{v_i}) + \sum_i \frac{x_i a_i}{RTv_i} + \sum_i x_i ln\frac{v_i}{v_i} + \frac{Pv^E}{RT} \tag{17}$$

If we for the $b$-parameter in equation (16) use the "normal" mixing rule

$$b = \sum_i x_i b_i \tag{18}$$

then a mixing rule for the $a$-parameter may in principle be derived from assuming:

$$\frac{G^E}{RT}|_{EOS} = \frac{G^E}{RT}|_{G^E-model} \tag{19}$$

Equation (19) is assumed to hold at the composition, temperature and pressure of the system. In UNIWAALS (Gupte et al., 1986 and Gani et al., 1989), the van der Waals equation of state is used to evaluate $G^E/RT|_{EOS}$, and the modified UNIFAC model is - due to its superior temperature dependence - used to represent the $G^E/RT|_{G^E-model}$.

Sample results of UNIWAALS are shown in Figures 7 and 8.

An advantage of UNIWAALS is that the existing modified UNIFAC parameters may be used to evaluate the $a$-parameter in equation (16), thus ensuring equivalence between the EOS and the $G^E$-model predictions at low pressures, which at the same time - and with the same parameters - enable good prediction at high pressures. In this way the pressure limitation of the GAMMA-PHI approach has been removed, and UNIFAC has been extended to high presures.

Another advantage of UNIWAALS is that high pressure gas solubility may be "added" to the modified UNIFAC by merely determining gas-gas and gas-solvent group-interaction parameters. As indicated in Figure 9, the solvent group-interaction parameters are identical to those already determined from low pressure VLE data.

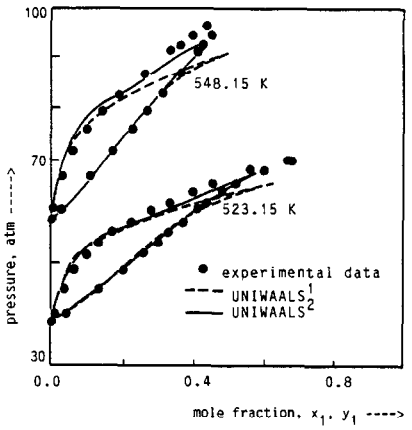



**Figure 7.** Comparison of VLE prediction with UNIWAALS and experimental data for the system, ethanol(1)-water(2) at 523.25 K and 548.15 K (Gmehling, J. and Onken, U.,"Vapor-Liquid Equilibrium Data Collection", DECHEMA Chemistry Data Series. Frankfurt, 1977).

Note: UNIWAALS[1] (results from Gupte et al. (1986))

UNIWAALS[2] ( results from Gani et al. (1989))

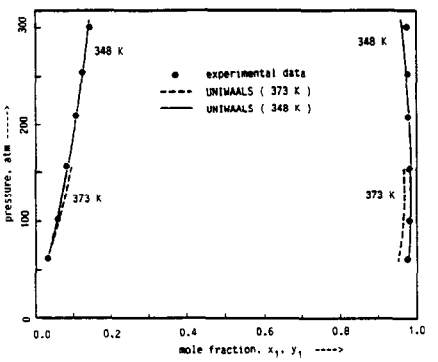



**Figure 8.** Correlation of VLE experimental data with UNIWAALS for the system, nitrogen(1)-benzene(2) at 348 K and 373 K.

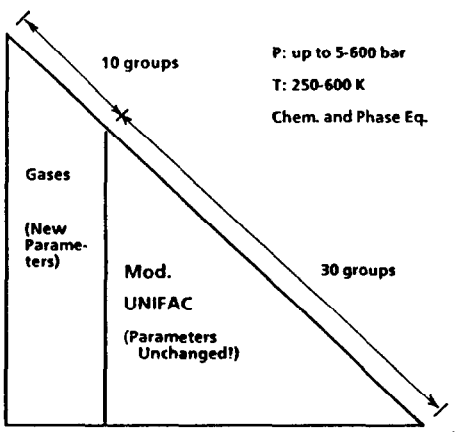



**Figure 9.** Solvent group and gas interaction parameters for high pressure application of UNIFAC via the UNIWAALS EOS.

An alternative approach to UNIWAALS is to generate artificial vapor-liquid equilibrium "data points" using UNIFAC and then fit parameters in an EOS to these points. Examples of such a "model unification" are given by Castier (1988) and Renon et al. (1989).

## MIXTURES WITH POLYMERS

Estimation of solvent activities in polymer plus solvent solutions may be done using either the activity coefficient approach, equation (3) or the EOS approach, equation (4). In the activity coefficient approach, the starting point is that of Flory and Huggins, equation (5). In the EOS approach the starting point is normally the equation of Flory (1970).

### The Polymer Activity Coefficient Approach

Previous attempts to estimate solvent activities by use of the original UNIFAC model have in general resulted in an underestimation of the solvent activity. This has been explained by the fact that the UNIFAC model does not take into account differences in free volumes.

To compensate for this a "free-volume activity term" has been added to the combinatorial and residual activity coefficient terms (Oishi and Prausnitz, 1978).

$$ln\,\gamma_i = ln\,\gamma_i^C + ln\,\gamma_i^R + ln\,\gamma_i^{fv} \tag{20}$$

where ln $\gamma^{fv}$ is calculated by means of Flory's expression, and ln $\gamma_i^C$ and $\ln_i^R$ from UNIFAC. The expression for ln $\gamma_i^{fv}$ is:

$$ln\,\gamma_i^{fv} = 3c_i\,ln\left[\frac{v_i^{1/3}-1}{v_m^{1/3}-1}\right] - c_1\left[\left(\frac{v_i}{v_m}-1\right)\left(1-\frac{1}{v_i^{1/3}}\right)^{-1}\right] \tag{21}$$

$\tilde{v}_i$ is the reduced volume of the solvent = $v_i/v_i^*$.
$\tilde{v}_m$ is the reduced volume of the mixture = $v_m/v_m^*$.
$3c_i$ is the external degree of freedom per solvent molecule. $c_i$ is set to 1.1.

The hard-core volumes $v_i^*$ and $v_m^*$ are calculated from van der Waals volumes. The free-volume term, equation (21), has been found highly dependent on the values of the pure-component densities employed.

Elbro et al. (1988) suggest that equation (20) is replaced by:

$$ln\,\gamma_i = ln\,\gamma_i^{ent} + ln\,\gamma_i^R \tag{22}$$

This equation also needs accurate pure-component densities.

The residual term, $ln\,\gamma_i^R$, is the UNIQUAC expression with parameters evaluated from VLE data for mixtures of low molar mass components. For example, ethyl benzene is very nearly equal to the repeat unit (group) of polystyrene. Elbro et al. (1988) used VLE data for ethyl benzene plus the relevant solvents to evaluate the UNIQUAC parameters of the $ln\,\gamma_i^R$ in equation (22). Equation (9) was used to evaluate $ln\,\gamma_i^{ent}$. As shown in table 2 below, their results compare well with those of Oishi and Prausnitz (1978) and those of the Holten-Andersen equation of state (1987, see the following section).

| System: Polystyrene + : | T K | Elbro et al. | Holten-Andersen | Oshi and Prausnitz |
|---|---|---|---|---|
| Cyclohexane | 303 | 2.7 | 8.7 | 5.5 |
| Toluene | 298 | 0.9 | 0.9 | 5.1 |
| MEK | 283 | 7.2 | 5.0 | 4.3 |
| $CCl_4$ | 293 | 3.0 | n.a. | 4.9 |

**Table 2.** Mean percent deviations between calculated and experimental solvent activities for polystyrene solutions with energy interactions.

Figure 4 shows that aqueous polymer solutions and polydimethylsiloxane (PDMS) solutions both have components with similar free volume percentages, and consequently free volume considerations are not necessary for these systems.This has been demonstrated by Rasmussen and Rasmussen (1989), who used the UNIFAC model without any free volume corrections to predict the activity of water in various polymer solutions.

The Polymer Equation of State Approach

The model by Oishi and Prausnitz (1978) may be applied to a relatively wide range of mixtures due to the large number of UNIFAC parameters available.

Holten-Andersen's (1987) and other EOS for mixtures with polymers tend to be more accurate than that of Oishi and Prausnitz (see High and Danner, 1988), but due to relatively few group parameters being available, they are less generally applicable. Holten-Andersen's EOS is based on the equation-of-state approach of Flory (1970). Below is given a simplified version of Holten-Andersen's EOS (Dahl, 1989).

Following Flory, the equation of state may be given in terms of a free volume and an attractive contribution:

$$P = \frac{nRT}{V}\left(\frac{v^{1/3}+C}{v^{1/3}-1}\right) + \frac{E}{nV^*v} \qquad (23)$$

*Mod. Flory free volume* *attractive*

with

$$C = \sum_i n_i C_i / n$$

$$V^* = \sum_i n_i V^*_i / n$$

$E$ is given by

$$E = \sum_i \tfrac{1}{2} z q_i n_i \left\{ \varepsilon_{ii} + \frac{\sum_j \theta_j exp(-\Delta a_{ji}/RT)\Delta\varepsilon_{ji}}{\sum_k \theta_k exp(-\Delta a_{ki}/RT)} \right\} \qquad (24)$$

$n$ is the number of moles, and $v$ the reduced volume, $V$ is the total volume, $V^*$ the total hard core volume, and $C$ is a molecular "degree-of-freedom-parameter".

The attractive term is a density dependent UNIFAC-like term with interaction parameters:

$$\Delta\varepsilon_{ji} = \varepsilon_{ji} - \varepsilon_{ii}$$

For systems with no hydrogen bonding, $a_{ji}$ is given by

$$a_{ji} = \varepsilon_{ji}/v \tag{25}$$

and

$$\Delta a_{ji} = a_{ji} - a_{ii}$$

In this expression, only one binary parameter ($a_{ji} = a_{ij}$) is used for systems with no hydrogen bonding.

For hydrogen bonding systems, however, one parameter is not enough to describe the systems adequately. It has been found that binary vapor-liquid equilibrium data may be correlated by introducing an extra binary entropic interaction parameter, $\Delta S^{hb}{}_{ji}$, into the residual term:

$$a_{ji} = (a_{ji} - a_{ii}) - T\Delta S^{hb}_{ji} \tag{26}$$

The parameters $\varepsilon_{ji}$ and $\Delta S^{hb}{}_{ji}$ are obtained from VLE data for mixtures with low molar mass components only.

From the expressions given above, all necessary thermodynamic quantities, including activity coefficients, may be derived.

The model requires the following molecular parameters:

$V_i^*$, $q_i$, $C_i$, $\varepsilon_{ji}$, and for hydrogen bonding systems $\Delta S^{hb}{}_{ji}$, $\Delta S^{hb}{}_{ji}$.

Group-contribution expressions for all of these parameters are developed.

This approach is more predictive than the activity coefficient + free volume approach in the sense that the pure-component volumes are included in the model and do not need to be supplied externally.

An example of the results obtained are shown in Table 3 for polyethylene oxide (PEO) plus hydrocarbon solvents.

| Mixture | Weight infinite dilution activity coefficient | |
|---|---|---|
| | exp | calc |
| Hexane-PEO | 33.4 | 29.7 |
| Heptane-PEO | 33.4 | 34.9 |
| Octane-PEO | 37.9 | 41.7 |
| Benzene-PEO | 3.8 | 4.6 |
| Toluene-PEO | 4.5 | 4.5 |
| Ethylbenzene-PEO | 5.2 | 6.0 |

**Table 3.** Weight infinite dilution activity coefficients of alkane solvents in polyethylene oxide (PEO, MW 4000).

An extensive parameter table for equations (23) and (24) is being developed.

## MIXTURES WITH ELECTROLYTES

For mixtures with electrolytes, useful new methods for practical phase equilibrium calculations have appeared in the last decade. The main part of the work in this field is, however, restricted to aqueous solutions, whereas systems of mixed solvents with salts, acids or bases have received only minor attention.

Phase equilibria of mixed solvent/salt systems are of interest in many practical contexts such as distillation of salt-containing liquids, liquid-liquid extraction of mixtures including salts, salt-precipitation from mixed solvents, etc. In all cases the presence of salt in the liquid phase may have substantial influence on the phase equilibria of the systems.

Two models have recently appeared within this area: Mock et al. (1986) and Sander et al. (1986).

The above-mentioned models describe the influence of ion on solvent species such as water, alcohols, ketones, etc. The ionic species are assumed not to be present in the vapor-phase, and the liquid-phase interactions are divided into long range ion-ion interactions and short range ion-ion, ion-solvent, and solvent-solvent interactions.

Mock et al. (1986) use the NRTL model to describe the short range forces, and the long range forces are neglected. The model uses only binary parameters, i.e. those usually associated with the NRTL model: Non-randomness factors, $\alpha_{ij}$, and energy parameters, $\tau_{ij}$,($i$ and $j$ may be solvent, cation or anion). The parameters are regressed from VLE data of mixed-solvent electrolyte systems.

In the model by Sander et al. (1986), the long range ion-ion interactions are described by the Debye-Hückel expression, and the short range ion-ion, ion-solvent and solvent-solvent interactions are described using a modified UNIQUAC expression, with its separate combinatorial and residual parts:

$$ln\gamma_n = ln\gamma_n^{D\text{-}H} + ln\gamma_n^{C} + ln\gamma_n^{R} \tag{27}$$

A term $\Psi_{kl}$ in the residual part contains the model parameters:

$$\Psi_{kl} = exp(-a_{kl}/T)$$
$$a_{kl} = u_{kl} - u_{ll} \tag{28}$$

where $k$ and $l$ may be an ion or a solvent species and $u_{kl}$ and $u_{ll}$ are interaction energy parameters. Note that in equation (28) ions are considered to be groups, whereas - only due to lack of experimental data - the solvents are molecular species.

The ion-ion and ion-solvent interaction energy parameters are assumed ion specific, e.g. a given cation-solvent interaction is independent of which anion the cation belongs to. For mixed solvent/electrolyte systems it has proven to be convenient to introduce concentration dependence in the interaction energy parameters between ion $i$ and solvent $m$, $u_{im}$ and $u_{mi}$. The following concentration dependence is used:

$$a_{im} = a^{*}_{im} + \theta_i \sum_{j \neq i} \delta_{ij'm}\theta_j \tag{29}$$

$$a_{mi} = a^{*}_{mi} + \theta_i \sum_{j \neq i} \delta_{ij'm}\theta_j \tag{30}$$

where $a_{im}^{*} = u_{im} - u_{mm}$ and $a_{mi}^{*} = u_{mi} - u_{ii}$. $\theta_i$ is the surface area fraction of ion $i$ and $\delta_{ij'm} = \delta_{ji'm}$ is a parameter which depends on the solvent $m$. The model parameters are thus $u_{ii}$, $u_{mm}$, and $u_{mi}$ and $\delta_{ij'm}$. In addition the pure ion and solvent volumes, and surfaces areas must be known, just as in the normal UNIQUAC model.

The above-mentioned mixed solvent/electrolyte models may be used for salt concentrations up to 10 molal. The salt effects are usually predicted very well. Sander et al. (1986) report an average deviation between predicted and experimental vapor-phase mole fraction for mixed solvent/electrolyte systems of 0.012. The ion activity coefficients may, however, not be well represented. The most

severe limitation of the models is lack of parameters due to scarcity of experimental information.

Figures 10 and 11 show two x-y diagrams predicted using the model by Sander et al. (1986). These diagrams are from a recent extension and modification of this model carried out by Macedo et al. (1989). The model now covers the following:

ions: $Li^+$, $Na^+$, $K^+$, $Ca^{2+}$, $Cu^{2+}$, $Ba^{2+}$, $Ni^{2+}$, $Hg^{2+}$, $Sr^{2+}$, $Cl^-$, $Br^-$, $NO_3^-$, $CH_3COO^-$, $F^-$, $I^-$

solvents: Water, methanol, ethanol, 1- and 2-propanol, 1-butanol, acetone

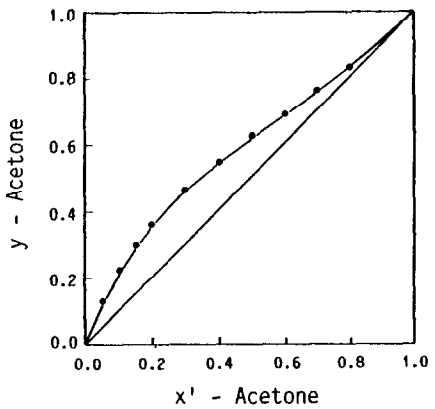



**Figure 10.** Experimental and calculated vapor-phase mole fraction y as a function of liquid phase salt free mole fraction x' for acetone/methanol/ 0.4742 Molal $CaCl_2$ at 1 atm.

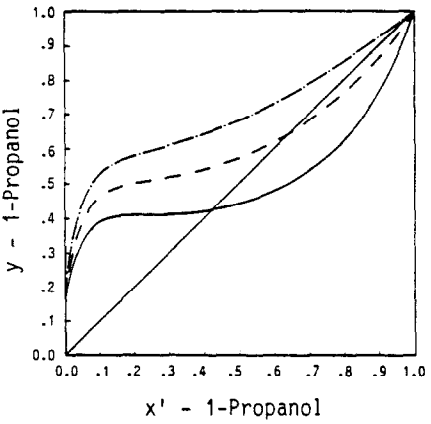



**Figure 11.** Vapor-phase mole fraction y as a function of liquid-phase salt free mole fraction x' for various salt concentrations in the system 1-propanol/$H_2O$/$SrCl_2$ at 80° C.

Rennotte (1987) has written an excellent review concerning models for mixtures with electrolytes, comparing the different models. Rennotte obtains improved results when he adds a Born contribution to the DTH activity coefficient expression, equation (27). This has been confirmed by Liu (1987).

It may be noted that the model by Sander et al. (1986), is formulated in terms of groups as far as the ions are concerned, whereas it is molecular for the solvents. This is because data for mixed solvent/electrolyte systems only have been measured for a limited number of solvents

## CONCLUSION

It has been shown that for many industrially important applications it is possible to accurately predict phase equilibria from knowing only the molecular structure. Indeed, the area of group-contribution predictions of phase equilibria has matured. The models have proven successful in numerous practical applications. The rate of addition of new (missing) group-interaction parameters to the UNIFAC parameter tables is today not limited by the effort needed in computing the parameters. Nearly all relevant data available in the literature have been correlated. Instead, the rate of addition of new UNIFAC parameters is limited by the rate at which new data are measured - which is a slow process!

During the next few years the challenges within this field will be to systematically extend group-contribution models to include ions and polymers and also to extend the models to high pressures and gas solubility by coupling these with equations of state.

Finally, it must be emphasized that the biggest limitation to further developing the models is not theoretical insight or computational shortcomings, but lack of good, relevant experimental phase equilibrium data.

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